# PROJECT BASED EXPERIENTIAL LEARNING REPORT

**ON**

## " Smart Rice Farming System "

## Under the Course

**PROJECT BASED EXPERIENTIAL LEARNING**

**SUBMITTED BY**


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# ABSTRACT

Agriculture plays an important role in food production and requires efficient management of water, environmental conditions, and crop health. Conventional irrigation and crop monitoring methods often depend on manual observation, which can result in unnecessary water consumption and delayed identification of crop diseases. To address these challenges, this project proposes an **AIoT-Based Smart Rice Farming System** that integrates IoT-based environmental monitoring, automated irrigation, and artificial intelligence-based rice disease detection.

The IoT component uses an **ESP8266 microcontroller** along with sensors such as a soil moisture sensor, rain sensor, DHT11 temperature and humidity sensor, LM35 temperature sensor, and LDR to monitor field conditions. Based on configured environmental conditions, the system can control an irrigation pump automatically and provide real-time field parameter information.

The AI component focuses on the detection of major rice diseases. A dataset of rice crop images was manually labelled for three major diseases, namely **Rice Blast, Rice Blight, and Brown Spot**. A YOLO-based object detection model was trained using the labelled dataset to identify the presence of these diseases in rice crop images. In addition, a machine learning-based approach is incorporated for **disease severity prediction and treatment/management recommendations** based on the detected disease and its severity.

The proposed system aims to integrate the IoT and AI components through a common software/dashboard layer. Sensor information and AI-based crop-health results can be used to provide a more comprehensive view of crop conditions. The proposed system can further be extended with automated image acquisition, historical data storage, weather information, multiple irrigation zones, and field-specific model improvement.

# 1. INTRODUCTION

The development of smart agriculture technologies has enabled the use of **Internet of Things (IoT), Artificial Intelligence (AI), machine learning, computer vision, and automation** for improving agricultural monitoring and resource management.

Rice is one of the major food crops and is affected by several diseases that can reduce crop productivity and quality. Diseases such as **Blast, Blight, and Brown Spot** can affect rice plants at different stages. Early identification of disease can help farmers inspect affected areas and take appropriate management measures.

At the same time, irrigation is one of the important aspects of crop management. Manual irrigation may result in over-irrigation or under-irrigation because the actual soil and environmental conditions may not always be considered.

The proposed project combines these two aspects into a single **AIoT-based Smart Rice Farming System**.

The IoT subsystem monitors parameters such as:

- Soil moisture
- Temperature
- Humidity
- Rain conditions
- Light intensity

The AI subsystem analyses rice crop images using a trained YOLO model to detect:

- Rice Blast
- Rice Blight
- Brown Spot

The detected disease information is further processed for **severity prediction and treatment/management recommendations**.

Thus, the proposed system aims to provide both **field-condition monitoring and AI-assisted crop disease monitoring** within a common smart agriculture platform.

## 2. PROBLEM STATEMENT

Traditional agricultural practices often depend on manual monitoring of soil conditions, irrigation requirements, and crop health. Such approaches can make continuous monitoring difficult and may delay the identification of crop diseases.

There is a need for an integrated system that can:

1. Monitor important environmental and soil parameters.
2. Automate irrigation based on field conditions.
3. Detects major rice crop diseases from images.
4. Estimate the severity of detected diseases.
5. Provide appropriate treatment or management recommendations.
6. Present the collected information through a unified monitoring platform.
7. Provide a scalable foundation for future automated crop-image acquisition and field-level monitoring.

Therefore, the project aims to develop an **AIoT-based smart rice farming system integrating IoT-based irrigation and AI-based disease detection, severity prediction, and management support.**

## 3. OBJECTIVES

The major objectives of the proposed project are:

1. To develop an IoT-based system for monitoring soil and environmental conditions in a rice crop environment.
2. To use an **ESP8266** for collecting sensor data and controlling the irrigation system.
3. To develop an automated irrigation mechanism using soil moisture and rainfall conditions along with other environmental parameters.
4. To manually label rice crop images for three major diseases:
    - Rice Blast
    - Rice Blight
    - Brown Spot
5. To train a **YOLO-based computer vision model** for detecting the selected rice diseases from crop images.
6. To develop a machine learning-based approach for **disease severity prediction**.
7. To provide disease-specific **treatment/management recommendations** based on the detected disease and predicted severity.
8. To integrate IoT sensor information and AI-based crop disease results through a common software/dashboard platform.
9. To maintain historical records of sensor readings and AI-based disease observations wherever applicable.
10. To design the system in a scalable manner so that future versions can include automated image acquisition, weather data, multiple field zones, and field-specific image datasets.

## 4. BRIEF LITERATURE REVIEW

Recent developments in smart agriculture have focused on the integration of IoT and AI technologies for crop monitoring, irrigation management, and disease detection.

IoT-based agricultural systems commonly use sensors to monitor soil moisture, temperature, humidity, rainfall, and other environmental parameters. Microcontrollers such as ESP8266 can collect sensor information and communicate the data to cloud or web-based platforms. Such systems can be used to automate irrigation based on predefined conditions.

Computer vision and deep learning have also become important techniques for crop disease detection. Object detection models such as the **YOLO family of models** can identify disease-related features from plant images and provide detection results with confidence values.

For the proposed project, rice crop images are labelled for three selected diseases—Blast, Blight, and Brown Spot—and used to train a YOLO-based detection model. The project further extends disease detection toward severity prediction and treatment/management recommendations.

The literature and existing systems indicate that combining IoT-based environmental monitoring with AI-based crop image analysis can provide a broader crop-management platform than either technology individually. The proposed work therefore focuses on integrating these components into a single smart rice farming system.

# 5. PROPOSED METHODOLOGY

The proposed system consists of two major technological components:

**A. IoT-Based Smart Irrigation and Environmental Monitoring**

**B. AI-Based Rice Disease Detection and Crop Health Analysis**

These components are integrated through a common software/dashboard layer.

## 5.1 IoT-Based Monitoring

The IoT subsystem uses an **ESP8266** as the main controller.

The following sensors are proposed:

| Sensor | Parameter monitored | Purpose |
|---|---|---|
| Soil Moisture Sensor | Soil moisture | Determine irrigation requirement |
| Rain Sensor | Rain condition | Avoid unnecessary irrigation during rain |
| DHT11 | Temperature and humidity | Environmental monitoring |
| LM35 | Temperature | Additional temperature measurement |
| LDR | Light intensity | Monitor light conditions |

The sensor readings are collected by the ESP8266 and processed according to predefined irrigation conditions.

### 5.2 Smart Irrigation

The irrigation subsystem consists of:

**ESP8266 → Relay Module → Water Pump**

A basic decision process can be implemented as:

Sensor Data
↓
Check Soil Moisture
↓
Check Rain Condition
↓
Consider Environmental Conditions
↓
Irrigation Decision
↓
Pump ON / OFF

For example, when the soil moisture is below a configured threshold and rain is not detected, irrigation can be activated. When sufficient moisture is reached or rain is detected, irrigation can be stopped or suspended.

The thresholds can be configured and calibrated according to the prototype setup rather than being treated as universal values for all rice fields.

### 5.3 Rice Disease Dataset Preparation

The AI subsystem uses rice crop images obtained from available datasets.

The images are manually labelled for three selected diseases:

1. **Rice Blast**
2. **Rice Blight**
3. **Brown Spot**

The labelled dataset is divided into appropriate training, validation, and testing subsets for model development and evaluation.

### 5.4 YOLO-Based Disease Detection

The labelled rice disease dataset is used to train a YOLO-based object detection model.

The basic workflow is:

Rice Crop Image
↓
Image Pre-processing
↓
YOLO Model
↓
Disease Detection
↓
Disease Class + Confidence
↓
Crop Health Result

The model is designed to detect the selected rice diseases from input crop images.

The performance of the model can be evaluated using suitable object-detection metrics such as:

- Precision
- Recall
- mAP
- Detection confidence

### 5.5 Disease Severity Prediction

After disease detection, the system further analyses the detected disease to estimate its severity.

The proposed workflow is:

Crop Image
↓
YOLO Disease Detection
↓
Detected Disease
↓
Severity Prediction Model
↓
Severity Level

The severity prediction component can classify the detected condition into configured severity categories, depending on the dataset and model developed by the team.

The severity result can then be displayed along with the disease detection result.

### 5.6 Treatment / Management Recommendation

Based on the detected disease and predicted severity, the system provides a corresponding treatment or management recommendation.

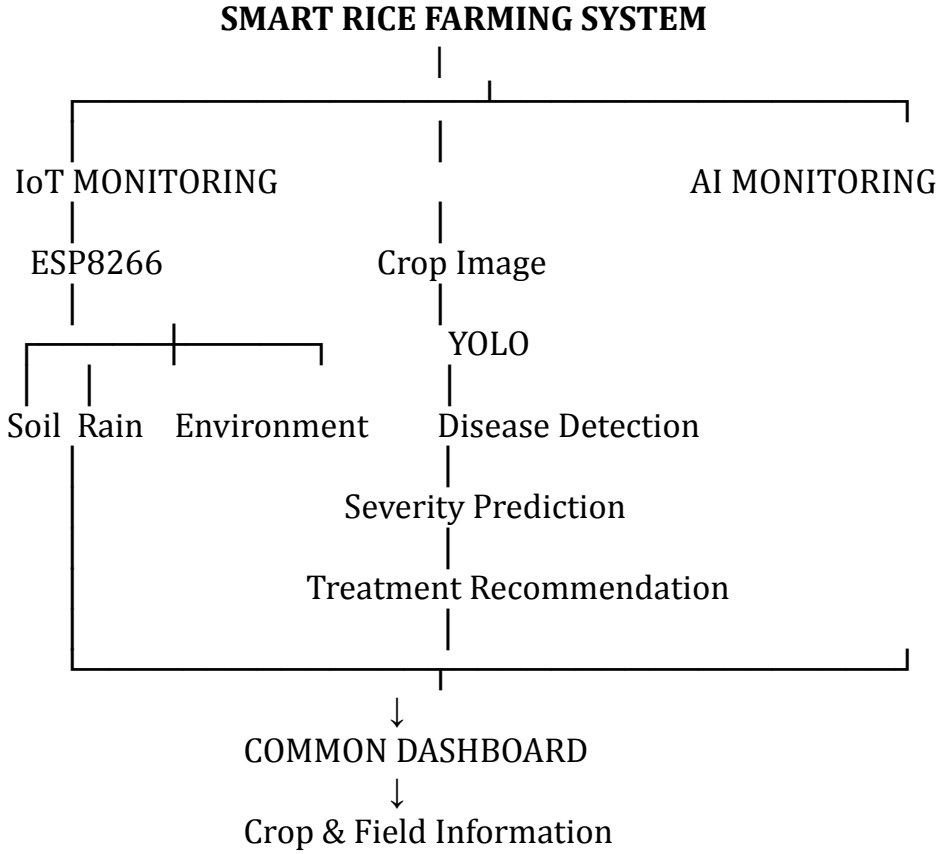
The workflow is:

Disease Detected
↓
Severity Prediction
↓
Recommendation Module
↓
Treatment / Management Guidance

The recommendations should be based on reliable agricultural references and presented as **decision-support information**, rather than replacing professional agricultural advice.

### 5.7 Integration of AI and IoT

The main proposed innovation of the project is the integration of both components.




The dashboard can display:

- Soil moisture
- Temperature
- Humidity
- Rain status
- Pump status
- Detected disease
- Detection confidence
- Predicted severity
- Treatment/management recommendation

### 5.8 Future Automated Image Acquisition

At the current stage, the YOLO model analyses available/input rice crop images and does not depend on continuous field imaging.

As a future extension, a camera such as an **ESP32-CAM or another suitable camera module** can be integrated to periodically capture crop images.

The future workflow can be:

Rice Plant
↓
Camera
↓
Periodic Image Capture
↓
Wi-Fi
↓
YOLOv8 Server / Processing System
↓
Disease Detection
↓
Database + Dashboard

This would allow the system to move toward automated periodic crop-health monitoring.

### 5.9 Future Data Integration

Future versions can store sensor and AI observations together:

Timestamp
+
Zone
+
Soil Moisture
+
Temperature
+
Humidity
+
Rain
+
Disease
+
Confidence
+
Severity

This can create a historical crop-health database and provide opportunities for future analytics and field-specific model improvement.

## 6. APPLICATIONS

The proposed system can have the following applications:

1. **Smart Rice Irrigation**
   Automated irrigation based on soil and environmental conditions.
2. **Rice Disease Detection**
   AI-assisted detection of selected rice diseases from crop images.
3. **Crop Health Monitoring**
   Combining environmental parameters with AI-based crop observations.
4. **Disease Severity Assessment**
   Providing an estimated severity category for detected diseases.
5. **Treatment/Management Decision Support**
   Providing disease-specific recommendations based on the detected condition and severity.
6. **Remote Farm Monitoring**
   IoT data can be made available through a suitable dashboard.
7. **Water Management**
   Monitoring soil moisture and controlling irrigation to reduce unnecessary watering.
8. **Agricultural Education and Research**
   The platform can be used as a prototype for studying AIoT applications in agriculture.
9. **Future Multi-Zone Agriculture**
   The system can be expanded to multiple agricultural zones with separate sensors and irrigation control.
10. **Future Automated Crop Imaging**
    Camera-based periodic crop-image acquisition can be integrated for automated disease monitoring.

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